

PROJECT REPORT

On

Budget Buddy : A Financial Calculator

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Under the Guidance of

Dr.Preeti Saini

Submitted By:

Rajveer Shaktawat 2210991362

Lavish Patiyal 2210991851

Neeraj Yadav 2210991970

Pushpender 2210992109

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Abstract/Keywords

The "Budget Buddy" project aims to provide a comprehensive financial management tool designed to help users manage their personal finances effectively. With a user-friendly interface developed using React.js, this web application allows users to track their income, expenses, and savings goals seamlessly. The backend is powered by Node.js and Express.js, ensuring robust and scalable performance. MongoDB is utilized as the database for efficient data storage and retrieval, allowing users to maintain detailed financial records. Key features of the application include user authentication, dynamic budget tracking, categorization of expenses, and generating insightful financial reports. The system employs JWT for secure user sessions, ensuring that sensitive financial data is protected. This project addresses the growing need for personal financial management tools by offering an accessible platform that can be customized to meet individual user needs. "Budget Buddy" not only facilitates day-to-day financial tracking but also provides future insights through budget analysis, helping users to set and achieve their financial goals. With the integration of modern web technologies, the project demonstrates a blend of technical innovation and practical utility, making it a valuable resource for anyone looking to manage their finances more effectively.

Keywords: Finance Calculator, MERN stack, Budget planning, Secure Data, Financial Report, Web Application, Saving Goals

1. Introduction :

In the realm of finance, your guide shall appear, "Budget Buddy" helps manage, with insight so clear. Track income and expenses, set goals with delight, Visualize your journey, from morning to night. Craft budgets with care, watch spending with ease, Secure and intuitive, your worries appease. With charts and reports, your path's understood, "Budget Buddy" makes managing finances feel good.

1.1 Background:

In the realm of personal finance where uncertainties abound, "Budget Buddy" emerges with solutions profound as daily transactions blend with financial dreams, this tool offers clarity beyond mere schemes. From managing your budget to tracking each expense, It brings structure and insight, offering recompense.

No longer must you navigate with confusion and strife, For "Budget Buddy" transforms your financial life. With a user-friendly interface and features robust, It turns budgeting into a practice of trust. Empowering you to plan, save, and thrive, It ensures your financial health is vibrant and alive.

Problem Statement:

In an era where financial missteps can lead to strife, Managing money often disrupts daily life. Budgets spiral out of control, and savings seem small, With expenses that fluctuate and debts that enthrall. Many face confusion with tracking their spend, Struggling to balance and make budgets mend. From hidden costs to unplanned needs that arise, The path to financial clarity often seems disguised.

"Budget Buddy" steps in to solve this pressing concern, Offering a structured approach for which many yearn. To streamline the process and make control precise, It's designed to turn financial woes into a strategy concise.

2. Software and Hardware Requirement Specification:

2.1 Methods:

The development of the Easy Trails application utilizes the MERN stack (MongoDB, Express.js, React.js, Node.js) for a full-stack solution. The project follows an agile development methodology, allowing for iterative progress and regular feedback. The application is built using a RESTful API architecture enabling seamless communication between the frontend and backend. Git is used for version control, ensuring efficient collaboration among team members.

2.2 Programming/Working Environment:

- Frontend: React.js is used for building the user interface, providing a dynamic and responsive experience. HTML, CSS, and JavaScript are used alongside React.js for styling and functionality.
- Backend: Node.js, along with Express.js, serves as the backend framework. It handles routing, middleware, and API requests.
- Database: MongoDB is used as the database, providing a flexible and scalable NoSQL environment to store user data, tour details, and reviews.
- Development Tools: Visual Studio Code (IDE), Git (Version Control), Postman (API testing), and GitHub (Repository hosting).

2.3 Requirements to Run the Application:

Hardware Requirements:

- Processor: Intel Core i3 or equivalent
- RAM: 4GB or higher
- Storage: Minimum 10GB of free disk space • Operating System: Windows 10, macOS, or Linux

Software Requirements:

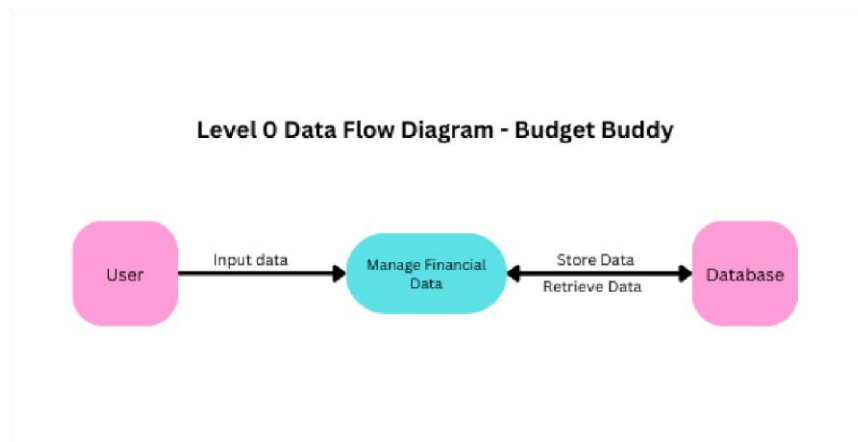
- Node.js (v14.x or higher)
- MongoDB (v4.x or higher)
- NPM (v6.x or higher)
- React.js (v17.x or higher)
- Browser: Google Chrome, Firefox, or any modern browser
- Text Editor: Visual Studio Code or any modern code editor



4. Database Analyzing, design and implementation

4.1 Data Flow Diagram Level 0

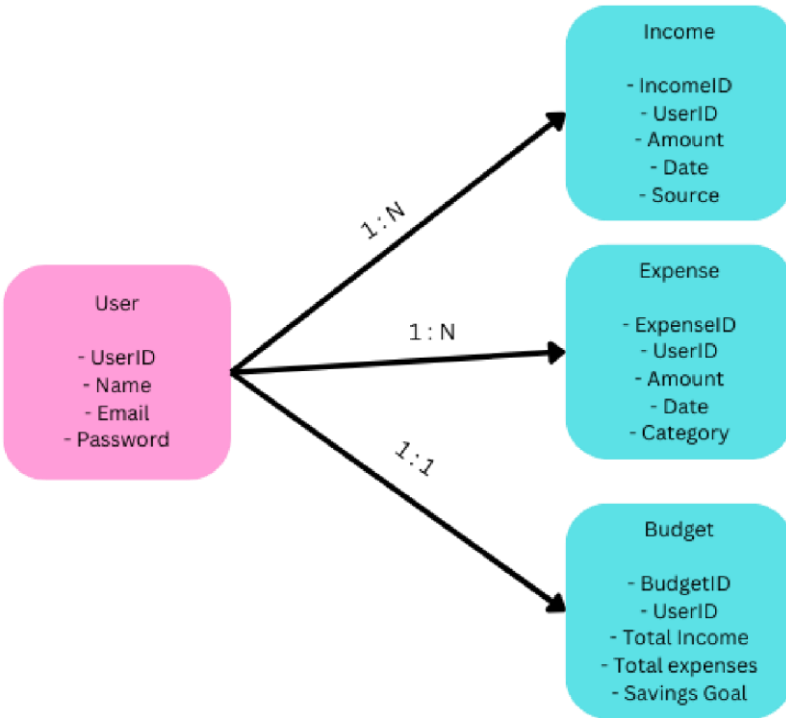
The Level 0 Data Flow Diagram (DFD) for the "Budget Buddy" project provides a high-level overview of the system, illustrating the interaction between the user and the Budget Buddy application. It shows the user as an external entity that inputs data such as registration details, income, and expenses, while the system processes this data to provide outputs like budget summaries and financial reports. This context diagram encapsulates the entire system as a single process interacting with the user.



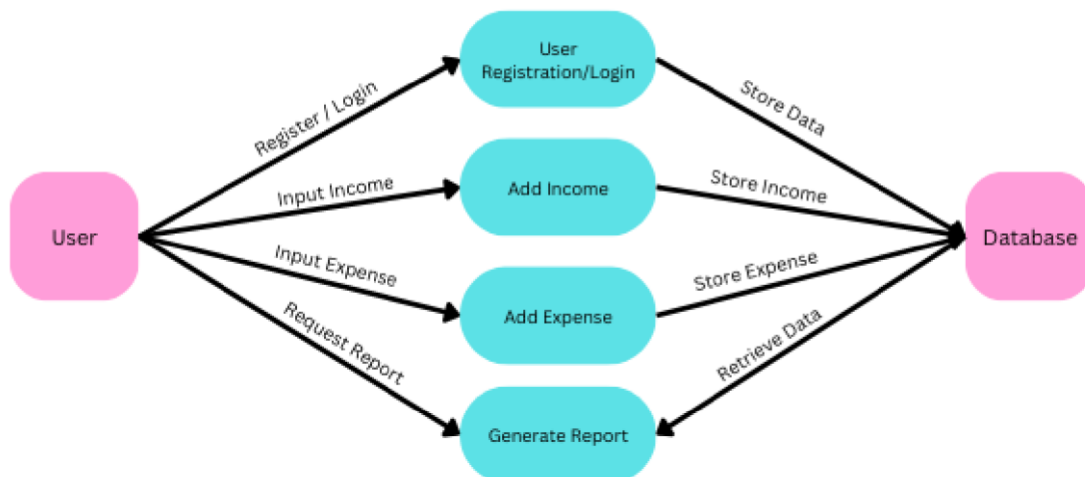
4.2 Data Flow Diagram Level 1

The Level 1 Data Flow Diagram (DFD) breaks down the main process into more detailed sub-processes, including User Registration/Login, Add Income, Add Expense, and Generate Report. Each of these processes interacts with the user and various data stores, such as User Data, Income Data, Expense Data, and Budget Data. The Level 1 DFD shows how data flows between these processes, ensuring that the system accurately manages and reports personal finances, from validating user credentials to storing and retrieving financial records for generating detailed reports.

Entity Relationship Diagram - Budegt Buddy



Level 1 Data Flow Diagram - Budget Buddy



4.3 Entity Relationship diagram

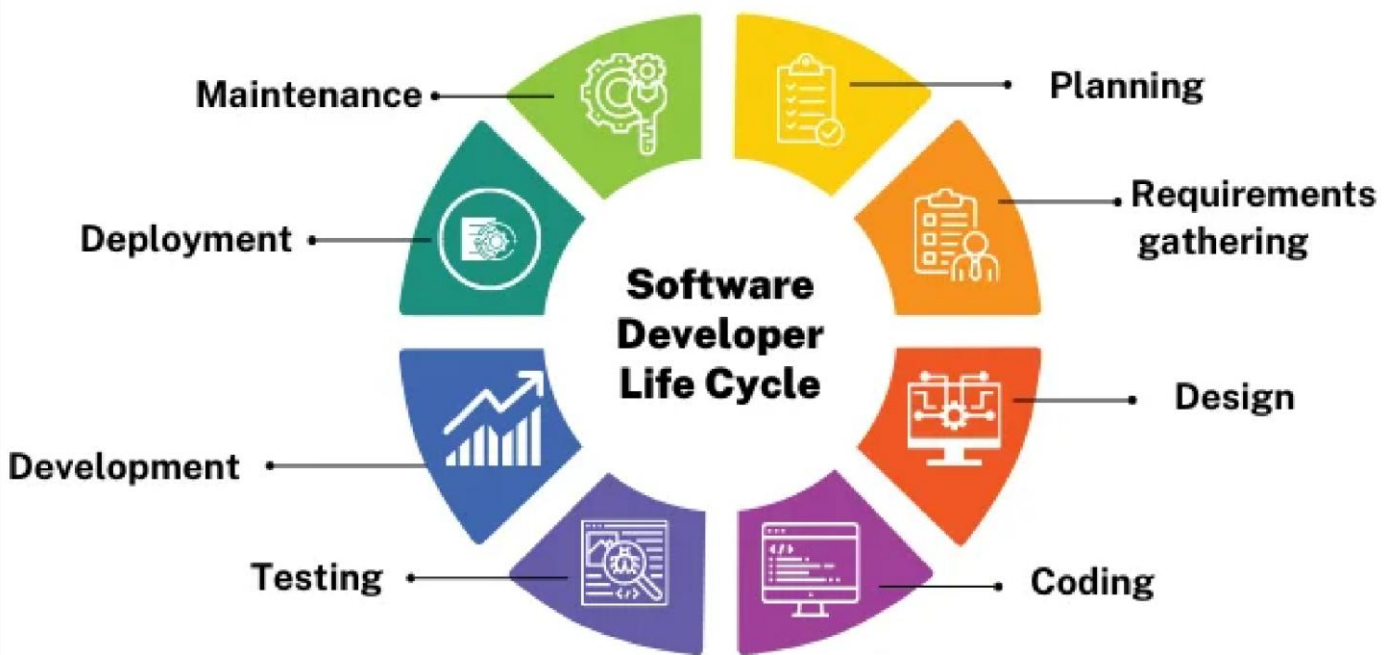
- Entities & their attributes:
 1. User: User ID, Name, Email, Password.
 2. Income: Income ID, User ID (Foreign Key), Amount, Date, Source.

3. Expense: Expense ID, User ID (Foreign Key), Amount, Date, Category.

4. Budget: Budget ID, User ID (Foreign Key), Total Income, Total Expenses, Savings Goal.

- Relationships:
- User to Income: One-to-Many (A user can have multiple income records).
- User to Expense: One-to-Many (A user can have multiple expense records).
- User to Budget: One-to-One (A user can have one budget plan).

4.4 Software Development Lifecycle(SDLC)



Budget Buddy follows the Agile SDLC approach, with the following phases:

- **Requirement Analysis:** Gathering and analyzing user requirements.
- **Design:** Creating DFD, ERD, and wireframes for the project.
- **Development:** Coding the frontend (React.js) and backend (Node.js), integrating MongoDB for data storage.
- **Testing:** Performing unit testing, integration testing, and user acceptance testing.
- **Deployment:** Deploying the application on a cloud platform like Heroku or AWS.
- **Maintenance:** Regular updates based on user feedback, fixing bugs, and adding new features.

4.5 Database Design & Schema

The project uses MongoDB as its database due to its flexibility in handling unstructured data, scalability, and ease of integration with Node.js.

Table Structure:

Although MongoDB is NoSQL, the structure is akin to:

- **Users Collection:**
 - `_id` (ObjectId)
 - `Name` (String) ▸ `Email` (String) ▸ `Password` (String)
- **Income Collection:**
 - `_id` (ObjectId)
 - `UserId` (ObjectId, Foreign Key)
 - `Amount` (Number)
 - `Date` (Date) ▸ `Source` (String)
- **Expense Collection:**
 - `_id` (ObjectId)
 - `UserId` (ObjectId, Foreign Key)
 - `Amount` (Number)
 - `Date` (Date) ▸ `Category` (String)
- **Budget Collection:**
 - `_id` (ObjectId)
 - `UserId` (ObjectId, Foreign Key)
 - `TotalIncome` (Number)
 - `TotalExpenses` (Number)
 - `SavingsGoal` (Number)

5. Program's Structure Analyzing and GUI Constructing (Project Snapshots)



Welcome Back

Sign in to your Budget Buddy account

Email Address

Password

☐ Remember Me

Sign in

Don't have an account [Create One](#)

Rajveer Singh Sh...
rajveersingh97737@gmail...

Dashboard

Income

Expenses

Profile

Logout

Dashboard

Welcome Back

Total Balance

₹245,000

+₹195,000 this month

Monthly Income

₹350,000

+12.5% from last month

Monthly Expense

₹155,000

0% from last month

Saving Rate

56%

Excellent

Financial Overview (This Month)

Finance Dashboard

Track your income and expenses

Add funds with Stripe

Create a secure checkout session for a card payment.

25

Pay with Stripe

Daily

Weekly

Monthly

Recent Transactions

Transactions are stacked by date (newest first)

gift

3/23/2026 Entertainment

-₹80000

salary

3/23/2026 Salary

+₹200000

cloths

3/14/2026 Shopping

-₹50000

12

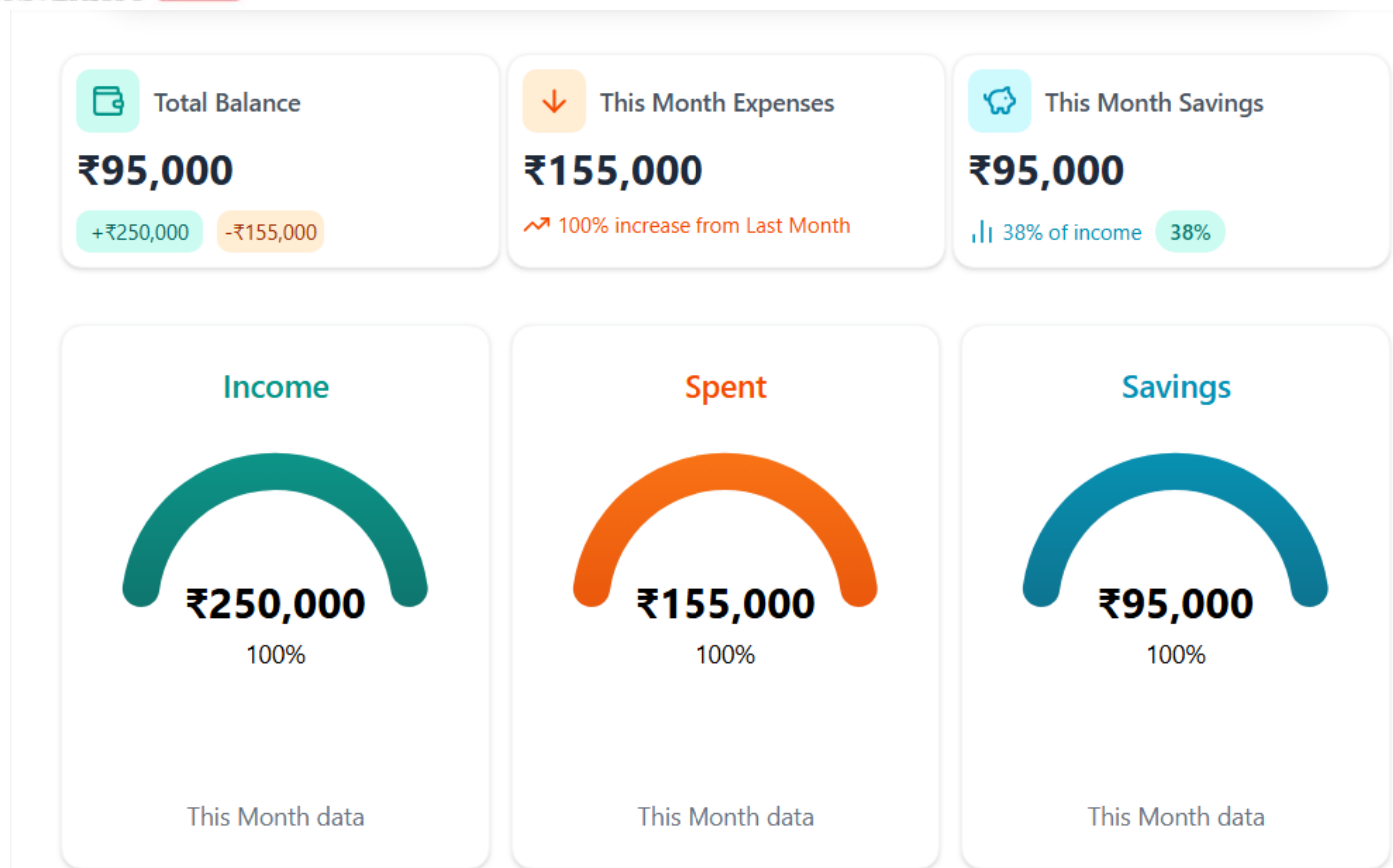


Fig 4. Add Transaction (Income / Expense)

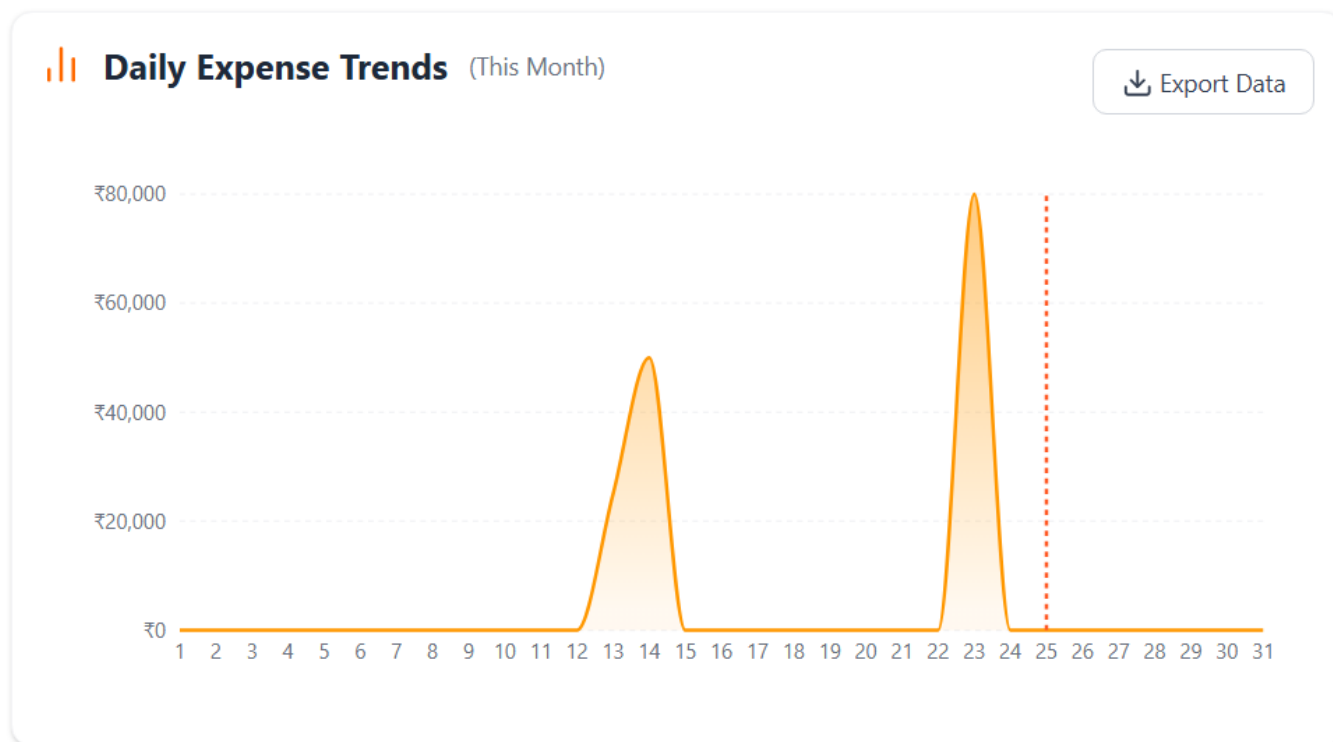
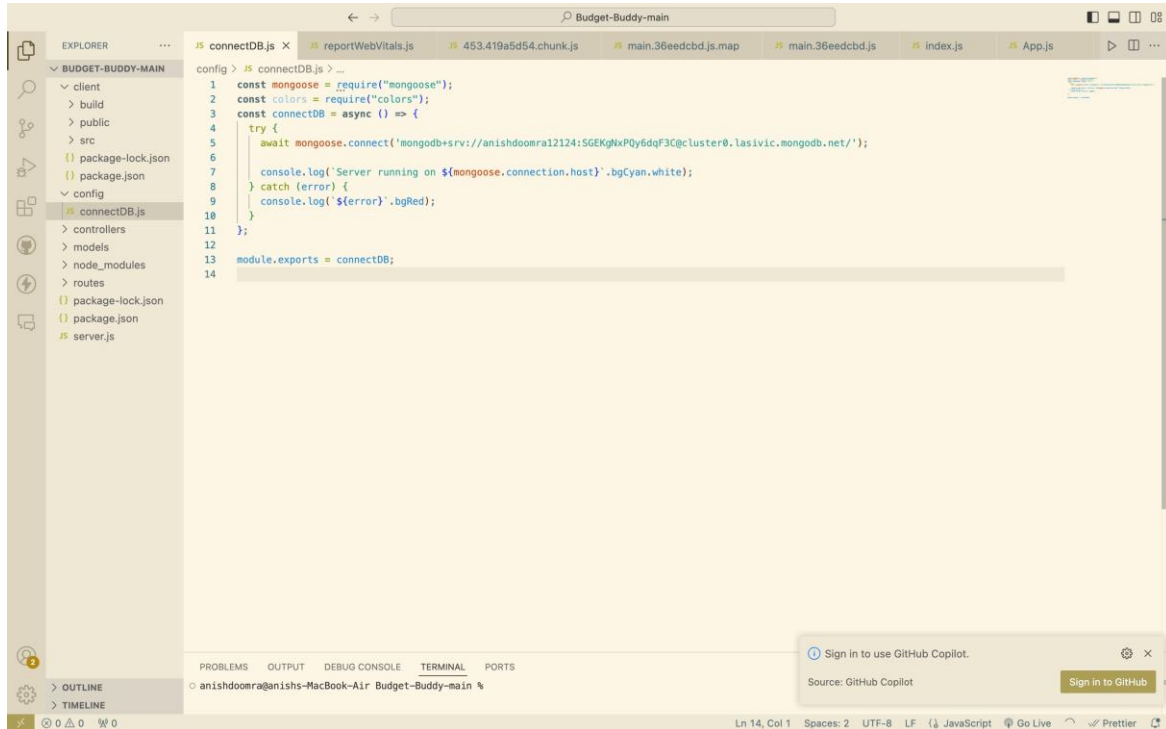


Fig 5. Expense Report

6. Code-Implementation and Database Connections

Database Transactions & Connectivity implementation:



```

1  const mongoose = require("mongoose");
2  const colors = require("colors");
3  const connectDB = async () => {
4    try {
5      await mongoose.connect(`mongodb+srv://anishdoomra12124:SGEKgNxPQy6dqF3C@cluster0.lasivic.mongodb.net/`);
6      console.log('Server running on $(mongoose.connection.host)'.bgCyan.white);
7    } catch (error) {
8      console.log(`${error}`.bgRed);
9    }
10   };
11
12   module.exports = connectDB;
13
14

```

Fig 6. Transactions stored on Database

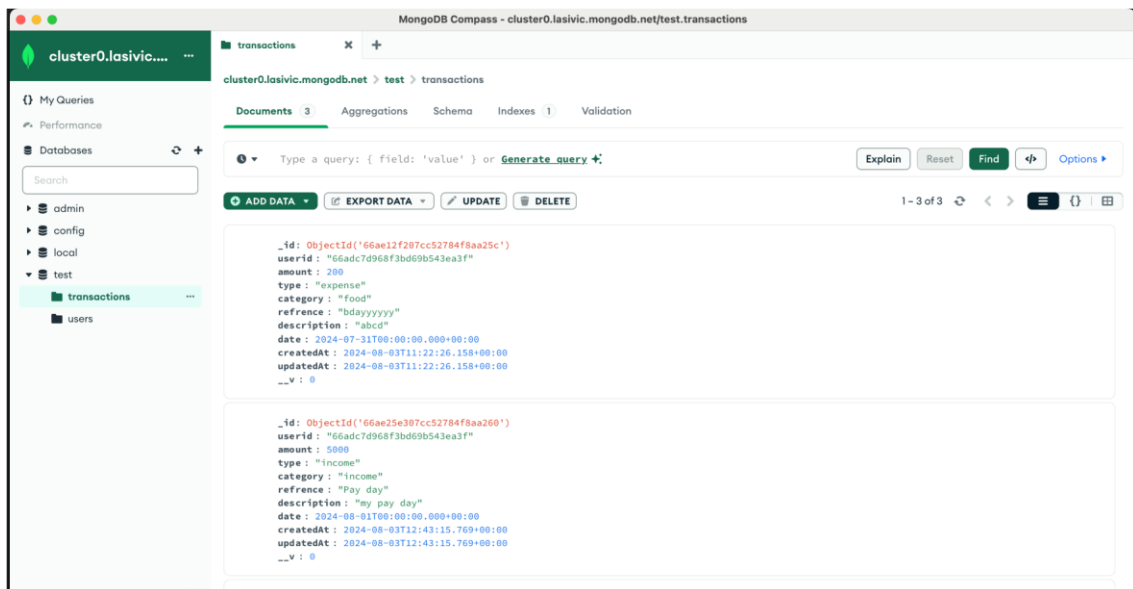


Fig 7. MongoDB connectivity

7. System Testing:

In testing phases, we delve deep and wide, Ensuring "Budget Buddy" performs with pride. From functional checks to user ease, We test each feature with meticulous precision and expertise. Performance and security are put to the test,

To ensure it functions optimally and meets the best. Compatibility and regression, we review with care, To guarantee a seamless experience everywhere. With each test, defects are tracked and resolved, Making sure our solution is robust and evolved.

8. Advantages & Disadvantages ▶ Advantages:

- Provides users with precise estimates of the amount needed for retirement by considering current expenses, inflation, and other financial variables.
- Offers an intuitive and user-friendly interface, making it easy for individuals to input their financial data and navigate through the portal.
- Provides actionable savings recommendations, guiding users on how much they should be saving each month to meet their retirement goals. ▶ Disadvantages:

- The accuracy of projections heavily relies on the accuracy of user inputs. Small errors in data entry can lead to significant discrepancies in retirement estimates.
- Storing sensitive financial information on an app or portal raises concerns about data security. Users may worry about the risk of unauthorized access or data breaches.

9. Conclusion :

The "Budget Buddy" project successfully addresses the need for an effective and user-friendly personal finance management tool. Through the development of this web application, users can easily track their income, expenses, and savings goals, gaining valuable insights into their financial habits. The use of modern web technologies like React.js, Node.js, Express.js, and MongoDB ensures that the application is not only efficient but also scalable and secure.

The project demonstrates the seamless integration of a robust back-end with an intuitive front-end, providing users with a smooth experience in managing their finances. By implementing features such as dynamic budget tracking, categorized expense management, and comprehensive financial reporting, "Budget Buddy" empowers users to take control of their financial well-being.

Looking ahead, the project has the potential for further enhancements, including advanced analytics, integration with financial institutions, and the incorporation of AI-driven financial advice. Overall, "Budget Buddy" serves as a solid foundation for a comprehensive financial management system, meeting the needs of users in today's fast-paced and increasingly digital world.

10. Future Scope

- Mobile Application: Extending the web application to a mobile platform.
- Integration with Banks: Connecting with bank APIs to auto-fetch transactions.
- AI-based Recommendations: Implementing AI to provide personalized financial advice.
- Multi-currency Support: Allowing users to manage finances in multiple currencies.
- Investment Tracking: Adding features for tracking investments like stocks and mutual funds.

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